Application of PaddleOCR in Power Consumption Environment Text Recognition

Baigen Wang
Anqing Power Supply Company
State Grid Anhui Electric Power Company
Anqing, Anhui, China
Email: 1057926110@qq.com

Tianze Zhao
Institute of Advanced Technology
University of Science and Technology of China
Hefei, Anhui, China
Email: canyon_ztz@mail.ustc.edu.cn

Lanfang Dong*
Institute of Advanced Technology
University of Science and Technology of China
Hefei, Anhui, China
*Email: lfdong@ustc.edu.cn

Zhaoyuan Yu
Anqing Power Supply Company
State Grid Anhui Electric Power Company
Anqing, Anhui, China
Email: lazyyzy@qq.com

Yalin Qian
Anqing Power Supply Company
State Grid Anhui Electric Power Company
Tongling, Anhui, China
Email: 1363297811@qq.com

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***Abstract*—In this paper, aiming at the demand of character recognition accuracy and efficiency in the electricity scene, An intelligent character recognition system based on Paddle OCR is proposed. The system uses PaddleOCRv3 version as the core model, combined with its pre-training advantages in industrial scenarios. Include multi-directional text processing, robustness to complex environments, and end-to-end optimization capabilities. By constructing a dedicated data set containing a specification banner and a power supply wiring diagram, and design post-processing mechanism such as confusing character map and pattern matching, It realizes the high-precision detection and identification of key information such as warning slogans and equipment numbers in the power scene. The experimental results show that on our self-built dataset of natural scenes with multiple lighting conditions and viewing angles, the system achieves an accuracy of 96% for object detection and 93.3% for recognition, which verifies the validity and practicability of PaddleOCRv3 in the task of character recognition in the complex environment of substation. It provides a feasible technical scheme for the intellectualization of power patrol inspection.**

Keywords—electricity scene, PaddleOCR, text detection, text recognition, confusion correction

I. INTRODUCTION

The intelligent development in the power scenario has put forward a clear demand for character recognition technology. Text information on the equipment or wall, such as warning signs, equipment numbers, parameter descriptions, etc. It is of great significance for the safe operation and maintenance management of the equipment. Therefore, character recognition requires high accuracy to ensure that the information is error-free.

Since the character recognition technology has entered the stage of digital processing, Scholars have done a series of research on how to improve the efficiency and accuracy of text detection and recognition. Traditional character recognition methods include algorithms based on contour matching, Otsu algorithm, recognition technology based on stroke decomposition, etc. These techniques are only applicable when the background is simple and the number of characters is small. The introduction of deep learning technology enables OCR systems to handle a variety of complex scenarios, including handwriting and multilingual recognition. Under the framework of in-depth learning, OCR processing is usually divided into three steps: image preprocessing, text detection and text recognition. There are also end-to-end learning methods. In 2015, CRNN architecture was proposed to realize end-to-end text recognition by combining CNN feature extraction and RNN sequence modeling[1].In 2017, the attention mechanism was proposed to be used in character recognition[2] to improve the accuracy of long text recognition. After 2018, the emergence of YOLO algorithm has made remarkable progress in text detection technology by dividing the input image into grids. Each grid unit directly predicts the coordinates of the bounding box, the confidence and the class probability of the target. Thereby completing the positioning of all the character areas[3] at one time, This design enables it to achieve extremely fast inference speed while maintaining high detection accuracy. It is far superior to traditional two-stage methods such as R-CNN series. PaddleOCR series models are proposed in 2020, covering text detection, text recognition, direction classification and other modules[4].End-to-end optimization. DB[5], SVTR and other algorithms are adopted, and complex scene enhancement technology is provided. Uch as multi-scale feature fusion and data enhancement strategies.

The characters in the electricity environment have the characteristics of high professionalism, complex environment and diverse shapes and directions. PaddleOCRv3 has been pre-trained on industrial scene data and contains a large number of fonts and parameter representations. Adaptable to high specialization with small data set tuning. Moreover, the training data contains a large number of synthetic severe condition data. The segmentation-based detection model DBNET is more robust to occlusion and other situations. In addition, PaddleOCR has a built-in directional classifier, which can detect and correct multi-directional text. It is more capable of recognizing curved text such as on the dashboard. In terms of model architecture, PaddleOCR adopts a unified framework of end-to-end optimization of “detection + recognition + directional” classification. The detection and recognition loss are combined with the back propagation, so that the feature correlation is higher, and the training efficiency is improved.

Therefore, in order to improve the efficiency of character recognition in the power consumption environment and reduce the cost of manual inspection, PaddleOCR series model[4] based on attention mechanism and convolutional neural network can be used for intelligent detection and recognition.

II. CONSTRUCTION OF PADDLEOCR

PaddleOCR is an open source OCR model launched by Baidu Paddle. After the continuous upgrading of versions v1, v2 and v3, the technical context is shown in Table I. The architecture always maintains the three-stage Pipeline design of "text detection + direction classification + text recognition". The development follows a clear industrial optimization path: under the strict inference speed limit, through systematic full-link optimization, Continuously improve accuracy.

TABLE I. PP-OCR VERSION COMPARISON

Version	Detection structure	Identify structure	Key contribution
PP-OCRv 1	MobileNet V3+FPN+ DB	MobileNetV3+ BiLSTM+CTC	Establish an ultra-lightweight baseline
PP-OCRv 2	MobileNet V3+FPN+ DB	PP-LCNet+BiL STM+CTC	CPPD distillation, U-DML mutual learning
PP-OCRv 3	MobileNet V3+RSE-F PN+DB	SVTR_LCNet+ GTC/CTC	LK-PAN/DML Teacher Model, RSE-FPN, SVTR and Six Lightweight Strategies

In this paper, PP-OCRv3 is selected as the core model, mainly based on the best balance between its technology maturity and scenario applicability. Version v3 introduces the SVTR lightweight text recognition network[6].The multi-scale feature fusion detection structure of PAN with large receptive field and residual attention mechanism has achieved a qualitative leap in accuracy. At the same time, it maintains the ultra-lightweight design and fully meets the dual requirements of complex font recognition and edge device deployment in substation scenarios. In addition, v3 has more abundant deployment cases and community support, and its technical solutions have been fully verified. It can ensure the reproducibility and engineering stability of the research results.

PP-OCRv3 is composed of three parts: detection, direction classification and recognition, all of which can be used independently. Each part has its own model and is trained using the PaddlePaddle framework. The model architecture is shown in Figure. 1 (the green box is the same as the v2 version, and the pink box is the new strategy of the v3 version).

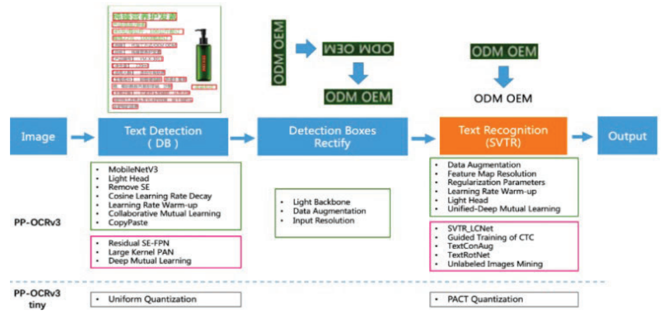

Figure. 1. PP-OCRv3 Model Architecture

In the text detection part, v3 improves the feature pyramid network for the student model. Residual SE-FPN network is proposed to enhance the important channel while preserving the residual link to avoid gradient disappearance[7].The optimization of the teacher model changes the convolution kernel in the path augmentation of the PAN structure from 3 * 3 to 9 * 9,A lightweight PAN-based structure LK-PAN with a larger receptive field was proposed[8].The DML distillation strategy is introduced to enable deep learning between student models from the feature representation level within the model[9].At the same time, it retains the cooperative mutual learning between CML tasks[10].

The recognition module of PP-OCRv3 is based on the optimization of the text recognition algorithm SVTR. SVTR no longer uses RNN structure. By introducing the Transformers structure, the context information[9] of the text line image is more effectively mined, O as to improve the text recognition capability. SVTR-LCNet is formed by fusing SVTR network and lightweight CNN network PP-LCNet, which greatly improves the inference speed.

The text orientation classifier in PaddleOCRv3 is an efficient, lightweight binary convolutional neural network. Its core design and implementation aim to achieve accurate orientation discrimination with minimal computational overhead[11]. Unlike integrated complex solutions like ABCNet, which addresses curved text detection and recognition holistically through adaptive Bezier curve networks[12], the classifier in PP-OCRv3 adopts a highly specialized and modular design approach. It utilizes a lightweight backbone network (such as the early stages of MobileNetV3 or a dedicated shallow CNN) as a feature extractor, receiving normalized text region images as input. At the end of the network, a global average pooling layer and a fully connected layer output a two-dimensional probability vector, corresponding to the confidence scores for the "0-degree" and "180-degree" classes, respectively. During the training phase, the model employs a standard cross-entropy loss function for end-to-end supervised learning on a large dataset containing pairs of normal and rotated text images.

Additionally, a set of five optimization strategies is introduced in PP-OCRv3[13], aimed at improving the capability of its recognition model.

A. GTC

GTC technique lies in constructing a collaborative learning framework between the attention mechanism and CTC. During training, the attention branch generates globally contextualized probability distributions through autoregressive decoding, producing soft-alignment supervisory signals for character sequences, while the CTC branch performs frame-wise classification predictions based on the same encoded features. The two are aligned via KL divergence loss, enabling CTC to retain its sequence-agnostic modeling advantage while assimilating contextual dependency features extracted by the attention mechanism. This multi-feature fusion approach inherits the attention mechanism's capacity for capturing long-range semantics while preserving the robustness and efficiency of CTC decoding. Consequently, during inference, only a lightweight single CTC branch needs to be deployed to achieve performance gains.

B. TextConAug

TextConAug is a data augmentation technique that enhances training data by mining textual contextual information. It operates by modeling contextual dependencies in natural language to perform semantically consistent word replacement at specific positions within a text. Based on pre-trained language models, the method substitutes target words with contextually appropriate alternatives sampled from the model's predicted probability distribution, prioritizing nouns, verbs, and other core semantic units. Through controlled temperature sampling, it balances augmentation diversity with grammatical coherence, effectively expanding the semantic boundaries of the training data. This approach not only increases the volume of training samples but, more importantly, deepens the model's understanding of linguistic structures and semantic relationships through context-aware lexical reorganization.

C. Collaborative mutual learning

The UDML collaborative mutual learning strategy enhances model performance by establishing a multi-branch co-supervision framework that implements knowledge distillation and feature alignment within the model. In the PP-OCRv3 architecture, this technique creates a three-tier joint supervision mechanism for the two heterogeneous network branches—SVTR_LCNet and Attention. Specifically, it enforces consistency constraints on the primary visual feature maps extracted by the PP-LCNet backbone, aligns the local contextual features from the SVTR module outputs, and coordinately optimizes the global semantic features at the Attention module output layer. This multi-granularity supervision enables deep fusion of the representation spaces while preserving the structural differences between the two branches: the SVTR branch enhances sensitivity to glyph structures, while the Attention branch improves sequence modeling capabilities. Ultimately, through gradient co-propagation, the overall model converges to a more generalized optimal state.

D. TextRotNet

TextRotNet is a self-supervised pretraining model that leverages large-scale unlabeled text line images. The core technique involves applying geometric transformations (such as rotation and perspective distortion) to text line images and training the model to predict the transformation parameters, thereby learning fundamental visual representations of textual structures. By designing pretext tasks like rotation angle prediction on massive unlabeled data, the convolutional neural network captures essential features including glyph morphology, stroke patterns, and spatial distribution characteristics of text. When transferred to downstream tasks, the pretrained weights from TextRotNet provide text-domain-optimized initialization for lightweight text recognition backbones like SVTR_LCNet. This approach mitigates overfitting in low-data scenarios and guides the model to converge to more favorable optima during fine-tuning, significantly enhancing recognition robustness under limited labeled resources.

E. UIM

UIM is a pseudo-label-based unlabeled data mining technique. First, a high-accuracy large-scale text recognition model is used to predict unlabeled images and generate confidence scores. Then, high-confidence predictions are selected as pseudo-labeled data through dynamic threshold filtering. Finally, the filtered reliable data is injected into the training process of the lightweight model, enabling efficient knowledge transfer from the large model to the small model and significantly improving the performance of the small model with limited annotated data.

III. APPLICATION

The data set comes from more than 200 national grid standard slogans and power supply line maps collected on the spot in the power consumption scene. PaddleLabel is used to label the data preliminarily, and then manual proofreading is carried out to ensure that the label is correct.

A. Experimental Setup

The hardware and software environment for the experiments in this paper is as follows: the CPU is an Intel i5-10300H at 2.50 GHz, the GPU is an NVIDIA GeForce GTX 1650 Ti with 4 GB of dedicated memory, the system RAM is 16 GB, the operating system is Windows x64, Python version is 3.8, CUDA version is 12.6, and PyTorch version is 1.12.1.

B. Model Setup

PaddleOCRv3 offers a comprehensive suite of pre-trained models, primarily comprising four major categories: text detection models (for localizing text regions, including the Chinese-English optimized ch_PP-OCRv3_det, the English-specific en_PP-OCRv3_det, and the multilingual-supporting ml_PP-OCRv3_det), text recognition models (for recognizing textual content, covering the core Chinese-English model ch_PP-OCRv3_rec, its high-precision distilled version ch_PP-OCRv3_rec_distill, the English-specific en_PP-OCRv3_rec, a digit recognition model, and multilingual models supporting over 80 languages such as korean_PP-OCRv3_rec and japan_PP-OCRv3_rec), text orientation classification models (e.g., ch_PP-OCRv3_cls for correcting text orientation), and end-to-end models (integrating detection and recognition, including the

Chinese-English ch_PP-OCRv3, English en_PP-OCRv3, and multilingual ml_PP-OCRv3 systems). This suite comprehensively addresses OCR application needs across different scenarios, languages, and accuracy requirements. Given that the power consumption environment involves mixed Chinese and English text along with special symbols, this study selects the Chinese-English end-to-end model ch_PP-OCRv3. By setting use_angle_cls=True, the orientation classifier is enabled to automatically correct text rotated at 0°/90°/180°/270°. The data is organized in the img_dir/label.txt format, with each line following the structure: image_path + text_content.

C. *Post-processing*

Aiming at the detection result under the power consumption scene, the method converts the detection result into atext line and corrects the numerical result in the result by using an established mode, Create a map of confusing characters, such as value confusion '0': [' O ',' o ',' ○'], unit confusion 'kV': ['kV', 'KV', 'Kv', 'K V'],Match the resulting text line according to a pattern such as' (\ d +) \ s * (kV | KV | Kv) 'and modify it if it matches the pattern and there is a corresponding confusion.

D. *Result*

Through the paddleOCRv3 model for testing, according to the demand in the power consumption scenario, the detection part is required to be as complete as possible. To prevent missing detection, the recognition part requires that the recognition of valid sentences should be completely accurate. Therefore, the test indicators are detection accuracy and recognition accuracy. Detection accuracy, i.e., the proportion of all labeled sentences to be detected that are actually detected; Recognition accuracy, that is, in all the marked sentences to be detected, The proportion of truly detected and completely matched sentences (ignoring blanks). The test data is shown in Table II.

TABLE II. TEST RESULTS OF AVERAGE DETECTION AND RECOGNITION ACCURACY AND REASONING TIME

Datasest	Average detection accuracy (%)	Average recognition accuracy (%)	Average inference time (ms)
Standardize slogans	96	93.3	68.2
Power supply circuit diagram	95.5	86.6	68.3

Visualization of the recognition results of the model is shown in Figures. 2 and 3. They respectively display the text lines detected in the dataset and the text detection results. By comparison, it can be observed that the key information in the circuit diagrams is correctly detected, and the detection results conform to the post-processing pattern. For keywords extraction, if the keywords "State Grid", "Wiring Diagram" and "Circuit Diagram" are detected, return Yes, otherwise return No.

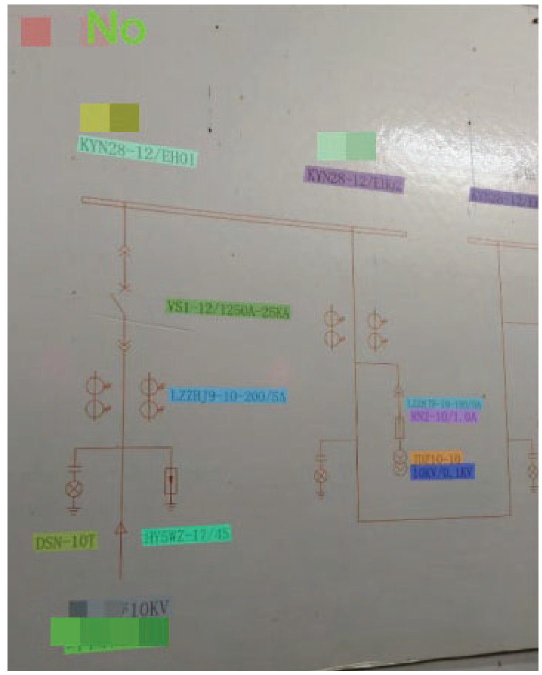

Figure. 2. Visualization of detected text locations

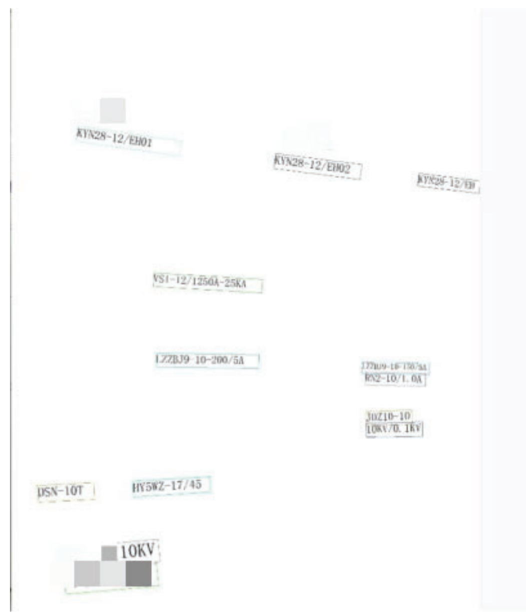

Figure. 3. Visualization of text recognition results.

IV. SUMMARY

Aiming at the urgent need of automatic management of standard identification at electricity customer sites, A standard text intelligent detection and recognition system based on in-depth learning is designed and implemented. The advanced PaddleOCRv3 model is systematically applied to the standard character recognition task in the complex power utility environments. By constructing a special standard slogan and a power supply line diagram data set and optimizing a model processing flow, A set of end-to-end intelligent recognition scheme is formed by integrating keyword extraction and result visualization post-processing modules. The test results show that the PP-OCRv3 model performs well in dealing with character recognition tasks in electricity usage environments. It achieves high detection and recognition accuracy on self-built data sets. The LK-pan structure of its detection module has excellent positioning ability for large size slogans. The SVTR architecture of the recognition module provides powerful modeling capabilities for complex fonts and contexts. It provides a feasible path for intelligent inspection and standardized management of electricity scene.

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